Core Potentials

Options for producing an all-electron-like density cube from ECP-based B3LYP and ZMP calculations

> **Key distinction.** An effective core potential (ECP) replaces explicit core electrons with an operator acting on valence electrons. The resulting orbitals yield an ECP valence (pseudo-)density; the ECP operator does not, by itself, specify a unique spatial distribution for the removed core electrons.

Why an ECP does not determine the missing density

An ECP is constructed to reproduce selected valence properties while eliminating explicit core degrees of freedom. Its radial channel potentials encode how valence states interact with the ion core. They are not occupied core orbitals or a tabulated core charge distribution. Distinct all-electron core densities can produce similar effective valence behavior, so the operator alone cannot be inverted into a unique density.

In this code path, PySCF constructs B3LYP and ZMP from the same molecule object, which carries its ECP. Their cube densities therefore already reflect the ECP through their orbitals. Adding a core term would be an explicit post-processing reconstruction model, not an ECP correction that PySCF can infer automatically.

Three possible reconstruction routes

Route	What is added	Strengths and limitations
All-electron reference	Run a compatible all-electron calculation and derive a correction, for example an all-electron minus ECP reference density.	System- and environment-aware. Requires an all-electron basis and calculation; a raw all-electron density cannot simply be added because it already contains valence density. Define and validate the correction carefully.
Fitted atomic-core model	Add element-specific atom-centered radial core-density functions, often as a sum over nuclei.	Fast and reusable; core-electron count and radial shape can be controlled. It is an approximation that may miss bonding/relaxation effects and needs a specified fit and validation set.
Dataset-specific reconstruction	Use matching pseudopotential metadata or an	Most systematic when the reconstruction data match the exact ECP, element, and

	augmentation/reconstruction dataset that supplies all-electron and pseudo partial waves and core information.	valence configuration. Not every ECP library supplies these data; an ECP potential alone is insufficient.

Density bookkeeping

For a chosen reconstruction, write the output density as

```
rho_reconstructed(r) = rho_ECP(r) + Delta rho_core(r)
```

Here, *Delta rho_core* must be defined by the chosen method. If it is intended to represent only the removed core electrons, its integral should equal the number of ECP-replaced electrons, subject to the convention and normalization of the selected pseudopotential. Reconstruction schemes can also include a correction to the valence density, so the correction's expected integral must be checked against that scheme rather than assumed.

For an all-electron reference, a possible definition is *Delta rho = rho_AE,ref - rho_ECP,ref*, evaluated with a consistent geometry, electronic state, grid, and density convention. Adding that difference to a different ECP density is still a modeling approximation and should be tested for transferability.

Applying it to B3LYP and ZMP cubes

1. Choose one reconstruction definition, parameter set, and ECP mapping for each element.
2. Evaluate the correction on the same spatial grid as each cube. Apply it separately to the B3LYP density and ZMP density; their valence densities differ, even though the same core model may be used.
3. Check integrated electron counts, non-negativity where expected, behavior near nuclei, and convergence with grid resolution. Compare against all-electron references for representative systems.
4. If cubes are used in gradient training, include the motion/derivative of atom-centered corrections consistently. Editing only `rho_cube_dft` (and its ZMP counterpart) does not update the SCF energy or nuclear gradients.

Practical recommendation

First check whether the exact ECP/basis data in use have a compatible density-reconstruction dataset. If so, use and validate that scheme. Otherwise, an all-electron reference-derived correction is a defensible research route when its cost and transferability are acceptable. A fitted atomic-core model is appropriate when speed and broad coverage matter more than environment-specific reconstruction, provided its parameters and validation are documented.

> **Do not add an arbitrary “core density” inferred from ECP coefficients.** Record the ECP name/version, reconstruction source, electron-count convention, grid procedure, and validation results with generated data.

Scope note: This document describes methodological choices; it does not select or implement a reconstruction model for the project.